FORMULARIO

TENSORE ENERGIA MOMENTO

$$\Pi^{\alpha} = \frac{\partial \mathcal{L}}{\partial(\partial_0 \phi_\alpha)}$$

$$T^{\mu\nu} = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_i)} \partial^\nu \phi_i - g^{\mu\nu}\mathcal{L}$$

Energy momentum tensor

$$\partial_\mu T^{\mu\nu} = 0$$

$$P^\nu = \int dx^3 T^{0\nu}$$

Conserved charge

ESEMPI IMPORTANTI

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + \frac{1}{2}(\partial_\mu \phi \partial^\mu \phi - m^2\phi^2) + \bar{\psi}(i\not{\partial} - m)\psi$$

$$\frac{\partial \mathcal{L}}{\partial(\partial_\mu A_\lambda)} = -F^{\mu\lambda}$$

$$\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} = \partial^\mu \phi$$

$$\frac{\partial \mathcal{L}}{\partial(\partial_\mu \psi)} = \bar{\psi} i \gamma^\mu$$

$$T^{\mu\nu} = -F^{\mu\lambda}\partial^\nu A_\lambda + \partial^\mu \phi \partial^\nu \phi + \bar{\psi} i \gamma^\mu \partial^\nu \psi - g^{\mu\nu}\mathcal{L}$$

IMP !! INDICE λ DIVERO E SOMMATO! UNO SU UNO SOTTO

$$\mathcal{H} = T^{00}$$

$$H = \int dx^3 \mathcal{H} = P^0$$

$$P^i = \int dx^3 T^{0i}$$

$$\mathcal{H} = \sum \Pi_i \dot{\phi}_i - \mathcal{L}$$

Eulero eq motto

$$\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} = 0$$

SIMMETRIE E CORRENTE / CARICA CONSERVATA

$$j^\mu_a = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_i)}(\partial_\nu \phi_i A^\nu_a - F_{i,a}) - \mathcal{L} A^\mu_a = -\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_i)} \frac{\delta \phi_i}{\delta \varepsilon^a} - \mathcal{L}\frac{\delta x^\mu}{\delta \varepsilon^a}$$

dove

$$\delta x^\mu = \varepsilon^a A^\mu_a$$

$$\phi'_i(x') - \phi_i(x) = \varepsilon^a F_{i,a}$$

$$\delta \phi_i = \phi'_i(x') - \phi_i(x) = \varepsilon^a (F_{ai} - A^\mu_a \partial_\mu \phi_i)$$

Se la simmetria è INTERNA $\delta x = 0$ cioè $A^\mu_0 = 0$ cioè

$$j^\mu_a = -\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_i)} \frac{\delta \phi_i}{\varepsilon^a}$$

$\Rightarrow \partial_\mu j^\mu = 0$ Leggi di conservazione

Carica conservata

$$Q_a = \int d^3x \, j^0_a$$

$\Rightarrow \partial_0 Q = 0$

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CAMPO SCALARE

$\mathcal{L} = \frac{1}{2}(\partial_\mu \phi \partial^\mu \phi - m^2 \phi^2)$ $\quad (\partial_\mu \partial^\mu + m^2)\phi = 0$ KG Equation

$\phi(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_p}} \left(a_{\vec{p}}\, e^{-ip\cdot x} + a^\dagger_{\vec{p}}\, e^{+ip\cdot x} \right)$ $\quad E^2 = \vec{p}^2 + m^2$ $\quad p_\mu p^\mu = m^2 = E_p^2 - \vec{p}^2$

$[\phi(t,\vec{x}), \pi(t,\vec{y})] = i\,\delta^3(\vec{x}-\vec{y})$

$[a_p, a^\dagger_{p'}] = (2\pi)^3 \delta^{(3)}(\vec{p}-\vec{p}')$ $\quad$ tutti gli altri zero $[\,,\,] = \phi$

$a_p |0\rangle = \phi$ $\qquad a^\dagger_{\vec{p}} |0\rangle = |\vec{p}\rangle \sqrt{2E_{\vec{p}}}$

NORMAL ORDERING

$\hat{P}^i = \int d^3x : T^{0i} : = \int \frac{d^3p}{(2\pi)^3} p^i\, a_p^\dagger a_p$

$: H : = \int \frac{d^3p}{(2\pi)^3} \frac{E_p}{2} : (a_p^\dagger a_p + a_p a_p^\dagger) : = \int \frac{d^3p}{(2\pi)^3} E_p\, a_p^\dagger a_p$

$: a_p a_{p'}^\dagger : = a_{p'}^\dagger a_p$ FOR SCALARS

CAMPO SCALARE COMPLESSO

$\mathcal{L} = \partial_\mu \phi^* \partial^\mu \phi - m^2 |\phi|^2$

$\phi(x) = \int \frac{d^3p}{(2\pi)^3 \sqrt{2E_p}} \left(a_{\vec{p}}\, e^{-ip\cdot x} + b^\dagger_{\vec{p}}\, e^{ip\cdot x} \right)$

$[a_p, a^\dagger_{p'}] = [b_p, b^\dagger_{p'}] = (2\pi)^3 \delta^{(3)}(\vec{p}-\vec{p}')$ $\quad$ tutti altri $[\,] = \phi$ nulli

$\hat{H} = \int \frac{d^3p}{(2\pi)^3} E_p \left(a_p^\dagger a_p + b_p^\dagger b_p \right)$

$\hat{P}^i = \int \frac{d^3p}{(2\pi)^3} p^i \left(a_p^\dagger a_p + b_p^\dagger b_p \right)$

$Q = \int \frac{d^3p}{(2\pi)^3} \left(a_p^\dagger a_p - b_p^\dagger b_p \right)$ $\quad$ n° di particelle − antiparticelle

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CAMPI FERMIONICI

WEYL

$\mathcal{L}_L = i\psi_L^\dagger \bar{\sigma}^\mu \partial_\mu \psi_L$ $\qquad$ $\mathcal{L}_R = i\psi_R^\dagger \sigma^\mu \partial_\mu \psi_R$

$(\partial_0 - \sigma^i \partial_i)\psi_L = 0$ $\qquad$ $(\partial_0 + \sigma^i \partial_i)\psi_R = 0$ $\qquad$ weyl equations

$\Vert$

PROPERTIES

$\sigma^2 \sigma^i \sigma^2 = -(\sigma^i)^*$ $\qquad$ $\sigma^i \sigma^j = \delta^{ij} + i\varepsilon^{ijk}\sigma^k$

DIRAC IN CHIRAL REPRESENTATION

$\mathcal{L} = \bar{\psi}(i\partial\!\!\!/ - m)\psi$ $\qquad$ $\psi = \begin{pmatrix}\psi_L \\ \psi_R\end{pmatrix}$ $\qquad$ $\bar{\psi} = \psi^\dagger \gamma^0 = (\psi_R^\dagger \psi_L^\dagger)$

$(i\partial\!\!\!/ - m)\psi = 0$ $\qquad$ $\partial\!\!\!/ = \gamma^\mu \partial_\mu = \partial_\mu \gamma^\mu$ $\rightarrow$ $\boxed{\not{p}\not{p} = p^2}$ MA $\boxed{\partial\!\!\!/ \cdot b\!\!\!/ \neq a \cdot b}$

$\boxed{a\!\!\!/ b\!\!\!/ + b\!\!\!/ a\!\!\!/ = 2(a\cdot b)}$

↙ DYOC equation ↘

$(\not{p} - m)U(p) = 0$ particle $\qquad$ $(\not{p} + m)V(p) = 0$ antiparticle

$\gamma^\mu = \begin{bmatrix} 0 & \sigma^\mu \\ \bar{\sigma}^\mu & 0 \end{bmatrix}$ $\qquad$ $\gamma^{\mu\dagger} = \gamma^0 \gamma^\mu \gamma^0$ $\qquad$ $\gamma^0\gamma^0 = \mathbb{1}$

$\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}\mathbb{1}$ $\qquad$ $\{\gamma^5, \gamma^\mu\} = 0 \rightarrow \gamma^\mu\gamma^5 = -\gamma^5\gamma^\mu$

$\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$ $\qquad$ $\gamma^5 = \begin{bmatrix} -\mathbb{1} & 0 \\ 0 & +\mathbb{1} \end{bmatrix}$ $\qquad$ $\gamma^{5\dagger} = \gamma^5$ $\qquad$ $\gamma^5\gamma^5 = \mathbb{1}$

TRACES PROPERTIES OF γ^μ AND γ^5

$\mathrm{Tr}[\gamma^\mu] = 0$

$\mathrm{Tr}[\gamma^\mu\gamma^\nu] = g^{\mu\nu}\,4$

$\rightarrow \mathrm{Tr}[\gamma^\alpha\gamma^\beta\gamma^\gamma\gamma^\delta] = 4[g^{\alpha\beta}g^{\gamma\delta} - g^{\alpha\gamma}g^{\beta\delta} + g^{\alpha\delta}g^{\beta\gamma}]$

$\mathrm{Tr}[\text{dispari } \gamma] = 0$

$\mathrm{Tr}[\gamma^5] = 0$

$\mathrm{Tr}[\gamma^\mu\gamma^\nu\gamma^5] = 0$

$\mathrm{Tr}[\text{odd } \gamma \cdot \gamma^5] = 0$

$\mathrm{Tr}[\gamma^\alpha\gamma^\beta\gamma^\gamma\gamma^\delta \cdot \gamma^5] = \begin{cases} 0 \\ \neq 0 \end{cases}$ if two indices are the same $\alpha\neq\beta\neq\gamma\neq\delta$

CONTRACTIONS OF γ

$\gamma^\mu\gamma_\mu = 4\mathbb{1}$

$\gamma^\mu\gamma^\alpha\gamma_\mu = -2\gamma^\alpha$

$\gamma^\mu\gamma^\alpha\gamma^\beta\gamma_\mu = 4g^{\alpha\beta}\mathbb{1}$

$\gamma^\mu \not{p}\gamma_\mu = -2\not{p} \rightarrow \gamma^\mu \not{p}_1 \not{p}_2 \gamma_\mu = 4(p_1\cdot p_2)$

$\gamma^\mu \not{p}_1 \gamma^\nu \not{p}_2 \gamma_\mu = -2\not{p}_2\gamma^\nu\not{p}_1$

$\boxed{\gamma^\mu\gamma^\alpha\gamma^\beta\gamma^\delta\gamma_\mu = -2\gamma^\delta\gamma^\beta\gamma^\alpha}$

$\boxed{(\gamma^\nu\gamma^5)^\dagger = \gamma^0(\gamma^\nu\gamma^5)\gamma^0}$

DIRAC EQUATIONS ON PLANE WAVES $\psi = U(p)e^{-ip\cdot x}$ particle

$\psi = V(p)e^{ip\cdot x}$ antiparticle

$(i\partial\!\!\!/ - m)\psi$ DYOC equation

→ particle → $(\not{p} - m)U = 0$ → $\bar{U}(\not{p} - m) = 0$

→ antipode → $(\not{p} + m)V = 0$ → $\bar{V}(\not{p} + m) = 0$

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SPINOR PROPERTIES

$$\sum_s u^s(p)\bar{u}^s(p) = \not{p} + m \qquad \sum_s v^s(p)\bar{v}^s(p) = \not{p} - m$$

$$\bar{u}^r(p)u^s(p) = 2m\delta^{rs} \qquad \bar{v}^r(p)v^s(p) = -2m\delta^{rs}$$

$$u^{\dagger r}(p)u^s(p) = 2E_p\delta^{rs} \qquad v^{\dagger r}(p)v^s(p) = +2E_p\delta^{rs}$$

$$\bar{u}^r(p)v^s(p) = \bar{v}^r(p)u^s(p) = 0$$

All that is at equal p

$$\bar{u}^r(p)\gamma^\mu u^s(p) = 2p^\mu\delta^{rs}$$

$$\bar{v}^r(p)\gamma^\mu v^s(p) = 2p^\mu\delta^{rs}$$

phase and chiral symmetry of MASSLESS DIRAC $\mathcal{L}_D = \bar{\psi} i\not{\partial}\psi$

$\mathcal{L}_D$ DIRAC is invariant under both phase $\psi \to e^{i\theta}\psi$ and chiral $\psi \to e^{i\theta\gamma^5}\psi$ transf.

$$j^\mu = \bar{\psi}\gamma^\mu\psi \qquad Q = \int d^3x\,\bar{\psi}\gamma^0\psi = \int d^3x\,\psi^\dagger\psi$$

phase current — phase charge — # particle − n° antiparticle

$$j^\mu_A = \bar{\psi}\gamma^\mu\gamma^5\psi \qquad Q_A = \int d^3x\,\bar{\psi}\gamma^0\gamma^5\psi = \int d^3x\,\psi^\dagger\gamma^5\psi$$

chiral current — chiral charge — n° of RH fermions − n° of LH fermions

Se $m \neq 0$ ho solo simmetria di fase e NON ho simm chirale

QUANTIZATION

$$\{\psi_a(t,\vec{x}), \psi_b^\dagger(t,\vec{y})\} = \delta^{(3)}(\vec{x}-\vec{y})\,\delta_{ab}$$

$$\psi(x) = \int \frac{d^3p}{(2\pi)^3}\frac{1}{\sqrt{2E_p}}\sum_{s=1,2}\left(a_{\vec{p},s}\,u^s_{(p)}e^{-ip\cdot x} + b^\dagger_{\vec{p}s}\,v^s_{(p)}e^{ip\cdot x}\right)$$

$$\psi^\dagger(x) = \int \frac{d^3p}{(2\pi)^3}\frac{1}{\sqrt{2E_p}}\sum_{s=1,2}\left(a^\dagger_{p,s}\,u^{\dagger s}_{(p)}e^{+ip\cdot x} + b_{p,s}\,v^{\dagger s}_{(p)}e^{-ip\cdot x}\right)$$

$$\bar{\psi} = \psi^\dagger\gamma^0 = \int \frac{d^3p}{(2\pi)^3}\frac{1}{\sqrt{2E_p}}\sum_{s=1,2}\left(a^\dagger_{ps}\,\bar{u}^s_{(p)}e^{ip\cdot x} + b_{ps}\,\bar{v}^s_{(p)}e^{-ip\cdot x}\right)$$

NORMAL ORDERING

$:a_{ps}a^\dagger_{ps}: = -a^\dagger_{ps}a_{ps}$

swappo e prendo un − sign!

$$\{a_{\vec{p}r}, a^\dagger_{\vec{q}s}\} = \{b_{pr}, b^\dagger_{qs}\} = (2\pi)^3\,\delta^{(3)}(\vec{p}-\vec{q})\,\delta_{rs}$$ tutti altri zero

$$H = \int \frac{d^3p}{(2\pi)^3}\sum_s E_p\left(a^\dagger_{ps}a_{ps} + b^\dagger_{ps}b_{ps}\right) \qquad Q = \int \frac{d^3p}{(2\pi)^3}\sum_s\left(a^\dagger_{ps}a_{ps} - b^\dagger_{ps}b_{ps}\right)$$

$$P^i = \int \frac{d^3p}{(2\pi)^3}\sum_s p^i\left(a^\dagger_{ps}a_{ps} + b^\dagger_{ps}b_{ps}\right)$$

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CAMPO FOTONE

$A^\mu = (\varphi, \vec{A})$ $\quad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ $\quad F_{\mu\nu} = \begin{pmatrix} 0 & E_x & E_y & E_z \\ -E_x & 0 & B_z & -B_y \\ -E_y & -B_z & 0 & B_x \\ -E_z & B_y & -B_x & 0 \end{pmatrix}$

$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu}$

$\partial_\mu F^{\mu\nu} = 0$ eq of motion

$H = \int d^3x \, \mathcal{H} = \int d^3x \, \frac{1}{2} (\vec{E}^2 + \vec{B}^2)$ $\quad P^i = \int d^3 T^{0i} = \int d^3x \, (\vec{E} \wedge \vec{B})^i$

Gauge invariance

$A_\mu \longrightarrow A_\mu + \partial_\mu g(x)$

Coulomb gauge

$A_0 \overset{!}{=} 0 \quad \vec{\nabla} \cdot \vec{A} \overset{!}{=} 0$

Lorentz gauge

$\partial_\mu A^\mu \overset{!}{=} 0$

QUANTIZATION IN COULOMB GAUGE

$$A_\mu = \int \frac{d^3p}{(2\pi)^3} \sum_{\lambda=1,2} \left(\varepsilon_\mu(\vec{p}\lambda) a_{p\lambda} e^{-ip\cdot x} + \varepsilon^*_\mu(\vec{p}\lambda) a^\dagger_{\vec{p},\lambda} e^{ip\cdot x} \right) \frac{1}{\sqrt{2\omega_p}}$$

$\omega_p = p^0 = |\vec{p}|$ $\quad p^\mu \varepsilon_\mu = 0$

$[a_{p\lambda}, a^\dagger_{p'\nu}] = (2\pi)^3 \delta^{(3)}(\vec{p} - \vec{p}') \, \delta_{\lambda\nu}$ tutti altri zero

$\hat{H} = \frac{1}{2} \int d^3x \, :(\vec{E}^2 + \vec{B}^2): = \int \frac{dp^3}{(2\pi)^3} \omega_p \sum_{\lambda=1}^{2} a^\dagger_{p\lambda} a_{p\lambda}$

$P^i = \int \frac{dp^3}{(2\pi)^3} \sum_\lambda p^i \left(a^\dagger_{p\lambda} a_{p\lambda} \right)$

$\vec{S} = \int d^3x \, (\vec{E} \times \vec{A})$ Coulomb gauge where $\partial_0 A_\mu = F_{0\mu}$, $\partial_0 A^i = -E^i$

$S^\mu = \frac{1}{2} \varepsilon^{ijk} S^{ij}$

Polarizzazione circolare:

$a^\dagger_{p\pm} = \frac{1}{\sqrt{2}} \left(a^\dagger_{p1} \mp i a^\dagger_{p2} \right)$ $\quad S^z \, a^\dagger_{p\pm} |0\rangle = \pm a^\dagger_{p\pm} |0\rangle$

$\varepsilon_{p\pm} = \frac{1}{\sqrt{2}} \left(\varepsilon_{p1} \pm i \varepsilon_{p2} \right)$ autostati di elicità!

PROPERTIES

$\sum_\lambda \varepsilon_{\mu\lambda}(\vec{p}) \, \varepsilon^*_{\nu\lambda}(\vec{p}) = -g_{\mu\nu}$ $\longleftarrow$ $(\varepsilon^\mu)^\dagger = \varepsilon_\mu$

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DELTA DEFINITION

$$\int d^q x\, e^{\pm ipx} = (2\pi)^q \delta(p)$$

$$\int d^q p\, e^{\pm ipx} = (2\pi)^q \delta(x)$$

SEMPRE VALIDE

TRASFORMATA DI FOURIER SU BRA/KET

può avere convenzioni diverse

$$|x\rangle = \int d^d p\, |p\rangle\langle p|x\rangle = \int d^d p\, |p\rangle \frac{e^{-ipx}}{(2\pi)^{d/2}}$$

$$|p\rangle = \int d^d x\, |x\rangle\langle x|p\rangle = \int d^d x\, |x\rangle \frac{e^{ixp}}{(2\pi)^{d/2}}$$

sue volume

Convenzione SIMMETRICA:

$$\langle x|p\rangle = \frac{e^{ixp}}{(2\pi)^{d/2}} \qquad \langle p|x\rangle = \frac{e^{-ipx}}{(2\pi)^{d/2}}$$

$$\int d^d x\, |x\rangle\langle x| = \mathbb{1} \qquad \int d^d p\, |p\rangle\langle p| = \mathbb{1}$$

$$|x\rangle = \int d^d p\, |p\rangle\langle p|x\rangle = \int d^d p\, |p\rangle\, e^{-ipx}$$

$$|p\rangle = \int d^d x\, |x\rangle\langle x|p\rangle = \int d^d x\, |x\rangle\, e^{ixp}$$

Convenzione NON SIMMETRICA sue volume

$$\langle x|p\rangle = e^{ixp} \qquad \langle p|x\rangle = \frac{e^{-ipx}}{(2\pi)^q}$$

$$\int dx\, |x\rangle\langle x| = \mathbb{1} \qquad \int dp\, |p\rangle\langle p| = \mathbb{1}$$

In ogni caso vale sempre:

$$\langle x|p\rangle \sim e^{ixp} \qquad \langle p|x\rangle \sim e^{-ipx}$$

(Anche se potrei usare anche la convenzione con i segni I opposti ... ma non si fa di solito)

NOTA = CONVENZIONE SU FUNZIONI DI GREEN

$$D(x) = \int \frac{d^q p}{(2\pi)^q} e^{-ipx} \tilde{D}(p)$$

$$D(p) = \int d^q x\, e^{ipx} D(x)$$

Convenzione non simmetrica sul volume

è una convenzione su $D(p)$ e $D(x)$ che sono FUNZIONI DI GREEN! NON sono funzioni in senso di funzioni d'onda! Altrimenti la convenzione sui segni sarebbe l'opposta perché

$$f(x) = \langle x|f\rangle = \int dp\, \langle x|p\rangle\langle p|f\rangle = \int \frac{dp}{(2\pi)^{d/2}} e^{ipx} \tilde{f}(p)$$

$$\tilde{f}(p) = \langle p|f\rangle = \int dx\, \langle p|x\rangle\langle x|f\rangle = \int \frac{dx}{(2\pi)^{d/2}} e^{-ipx} f(x)$$

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→ Scelta = sempre così, unica formula

→ trasformata Fourier = scelto di convenzione $\pm i$ dell'esponenziale
scelto di convenzione volume simetrico o no

No ogni "MONDO" può avere convenzioni diverse! Il mondo di QM dei Ket / BRA e delle funzioni d'onda ha convenzione $\langle x|p\rangle = \frac{e^{ixp}}{()^{3/2}}$ $\langle p|x\rangle = \frac{e^{-ixp}}{()^{3/2}}$ simetrico e con $+i$ per $\langle q|p\rangle$.

Il mondo delle funzioni di Green ha considero con convenzione NON SIMMETRICA sul volume ....

L'IMPORTANTE È MANTENERE LA COERENZA DENTRO OGNI MONDO
(Se mondi diversi usano convenzioni diverse non è un problema perché non si parlano!
.. con fourier trasformo $D(p)$ in $D(x)$.. non posso mai trasformare $D(p)$ in $|x\rangle$ .. sono Mondi diversi!)

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WICK'S THEOREM

SCALARS

normal ordering $: a^\dagger a : = a^\dagger a$

time ordering $T\{\phi(x)\phi(y)\} = \begin{cases} \phi(x)\phi(y) & x^0 > y^0 \\ \phi(y)\phi(x) & y^0 > x^0 \end{cases}$

$$D(x-y) = \overbrace{\phi(x)\phi(y)} = \langle 0|T\{\phi_I(x)\phi_I(y)\}|0\rangle = \begin{cases} [\phi^+(x), \phi^-(y)] & x^0 > y^0 \\ [\phi^+(y), \phi^-(x)] & y^0 > x^0 \end{cases}$$

Wick's theorem

$$T\{\phi_1 \dots \phi_n\} = :\phi_1 \phi_2 \dots \phi_n: + :\{\text{all possible contractions}\}:$$

$$\langle 0|T\{\phi_1 \dots \phi_n\}|0\rangle = \text{all contractions COMPLETE}$$

FERMIONS

normal ordering $: a a^\dagger : = - a^\dagger a$ goes a minus sign each swap | Aim: $a^\dagger$ primo di a

time ordering $T\{\psi(x)\bar{\psi}(y)\} = \begin{cases} \psi(x)\bar{\psi}(y) & x^0 > y^0 \\ -\bar{\psi}(y)\psi(x) & y^0 > x^0 \end{cases}$ goes a minus sign each swap | Aim: primo il tempo maggiore

contraction under normal ordering $: \bar{\psi}\psi\psi\bar{\psi}: = - :\psi\bar{\psi}\psi\bar{\psi}: \xrightarrow{\text{contract}} -\overbrace{\psi\bar{\psi}}\,\overbrace{\psi\bar{\psi}}$ goes a minus sign each swap | Aim: ψ primo di $\bar{\psi}$ prima contiguo solo $\overbrace{\psi\bar{\psi}}$

$$S(x-y) = \overbrace{\psi(x)\bar{\psi}(y)} = \langle 0|T\{\psi(x)\bar{\psi}(y)\}|0\rangle = \begin{cases} \{\psi^+(x), \bar{\psi}^-(y)\} & x^0 > y^0 \\ -\{\bar{\psi}^+(y), \psi^-(x)\} & y^0 > x^0 \end{cases}$$

NOTA

$\overbrace{\psi\psi} = \overbrace{\bar{\psi}\bar{\psi}} = 0$ $\quad \overbrace{\bar{\psi}\psi}$ Non ha senso!

Wick's theorem

$$T\{\psi_1\bar{\psi}_2\psi_3\bar{\psi}_4\psi_5\bar{\psi}_6 \dots\} = :\psi_1\bar{\psi}_2\psi_3\bar{\psi}_4 \dots : + :\{\text{all contractions}\}:$$ & ricordando di aggiungere un meno ad ogni swap necessario

$$\langle 0|T\{\psi\bar{\psi}\dots\}|0\rangle = \text{all complete contractions \& minus sign at each swap}$$

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BOSONS

normal ordering and time ordering as for scalars

$$D_{\mu\nu}(x-y) = \langle 0|T\{A_\mu(x) A_\nu(y)\}|0\rangle = \overbrace{A_\mu(x) A_\nu(y)} = \begin{cases} [A_\mu^+(x), A_\nu^-(y)] & x^0 > y^0 \\ [A_\nu^+(y), A_\mu^-(x)] & y^0 > x^0 \end{cases}$$

WICK'S THEOREM

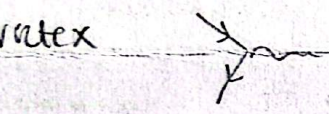
$$T\{A_{\mu_1} A_{\mu_2} \dots\} = :A_{\mu_1} A_{\mu_2} \dots: + :\{\text{All possible contractions}\}:$$

Feynmann rules for Bosons

- PROP $\tilde{D}(p) = \int d^4y\, e^{ipy} D_\mu(y) = \dfrac{-i\, g_{\mu\nu}}{p^2 + i\varepsilon}$

- EXT LEGS $\overbrace{A_\mu |p\lambda\rangle} = \varepsilon_\mu(p\lambda)\, e^{-ipx} |0\rangle$

$$\langle p\lambda | \overbrace{A_\mu} = \varepsilon^*_\mu(p\lambda)\, e^{ipx} \langle 0|$$

- QED vertex  $= -ie\,|Q|\,\gamma^\mu$

FROM

$H_I = e\bar{\psi} A_\mu \gamma^\mu \psi$

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LSZ

SHRODINGER PICTURE

$$|a(t)\rangle = e^{-iH(t-T_i)}\underbrace{|a(T_i)\rangle}_{=:|a\rangle}$$

$$|a(t_f)\rangle = e^{-iH(T_f-T_i)}|a\rangle$$

$$|b(t)\rangle = e^{-iH(t-T_i)}|b(T_i)\rangle$$

$$\underbrace{|b(T_f)\rangle}_{=:|b\rangle} = e^{-iH(T_f-T_i)}|b(T_i)\rangle$$

$$\langle b(T_f)|a(T_f)\rangle = \langle b(T_f)|e^{-iH(T_f-T_i)}|a(T_i)\rangle =: \langle b|e^{-iH(T_f-T_i)}|a\rangle$$

HEISEMBERG PICTURE

$$|a,T_i\rangle_H = e^{iHT_i}\underbrace{|a(T_i)\rangle_S}_{|a\rangle}$$

$$|b,T_f\rangle_H = e^{iHT_f}\underbrace{|b(T_f)\rangle_S}_{=|b\rangle}$$

$$\langle b,T_f|a,T_i\rangle_H = \langle b|e^{-iH(T_f-T_i)}|a\rangle$$

$$\langle b|S|a\rangle = \lim_{T_f-T_i\to\infty}\langle b,T_f|a,T_i\rangle = \lim_{T_f\to T_i\to\infty}\langle b|e^{-iH(T_f-T_i)}|a\rangle = (*)$$

which is equal to $(*) = \langle b|T\{e^{-i\int_{-\infty}^{\infty}dx^4 H_I(x)}\}|a\rangle$

↓

Hamiltoniano di interazione

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INTERACTING THEORIES

$$H = H_0 + H_{int} = \int d^3 \mathcal{H}_0 + \int dx^3 \mathcal{H}_{int}$$

$$\mathcal{L} = \mathcal{L}_0 + \mathcal{L}_{int}$$

if $\mathcal{L}_{int}$ does not contain derivatives of the fields we have $\mathcal{H}_{int} = -\mathcal{L}_{int}$

S-MATRIX

$$\langle b|S|a\rangle = \lim_{|T_f - T_i| \to \infty} \langle b| e^{-iH(T_f - T_i)} |a\rangle = \lim_{|P_f - T_i| \to \infty} \langle b, T_f | a, T_i\rangle$$

for:

$$S = \lim_{T_f - T_i \to \infty} e^{-iH(T_f - T_i)}$$

$$SS^\dagger = S^\dagger S = \mathbb{1} \qquad S = \mathbb{1} + iT$$

LSZ FORMULA

$$\langle k_1 \dots k_m | T_f | p_1 \dots p_n, T_i\rangle = \left(\frac{i}{\sqrt{Z}}\right)^{n+m} \prod_i^n \int dx_i^4 \, e^{-i p_i x_i} \prod_j^m \int dy_j^4 \, e^{+i k_j y_j} (\Box_{x_i} + m^2) \dots (\Box_{y_j} + m^2) \dots$$

(initial, final)

$$\cdot \langle \Omega | T\{\phi(x_1) \dots \phi(x_n)\phi(y_1) \dots \phi(y_m)\} | \Omega\rangle$$

for $p_1 + p_2 + \dots p_n \longrightarrow k_1 + k_2 \dots k_m$ process

$$\prod_i^n \int dx_i^4 \, e^{i p_i x_i} \prod_j^m \int dy_j^4 \, e^{-i k_j y_j} \langle \Omega | T\{\phi(x_1) \dots \phi(y_m)\} | \Omega\rangle = \prod_{r=1}^n \frac{(i\sqrt{Z})}{(p_r^2 - m^2)} \prod_{s=1}^m \frac{(i\sqrt{Z})}{(k_s^2 - m^2)} \langle k_1 \dots | T | p_1 \dots \rangle$$

(initial, final, initial, final)

for GREEN FUNCTION:

$$(\Box_{x_i} + m^2) \langle \Omega | T\{\phi(x_1) \dots \phi(y_m)\} | \Omega\rangle = -(p_i^2 - m^2) \langle \Omega | \dots | \Omega\rangle$$

$$(\Box_{y_j} + m^2) \langle \Omega | \dots | \Omega\rangle = -(k_j^2 - m^2) \langle \Omega | \dots | \Omega\rangle$$

$G(x_1 \dots x_n y_1 \dots y_m) := \langle \Omega | T\{\phi(x_1) \dots \phi(x_n)\phi(y_1) \dots \phi(y_m)\} | \Omega\rangle$ such that Fourier transform gives:

$$G(x) = \int \frac{dp^4}{(2\pi)^4} e^{-i p \cdot x} \tilde{G}(p) \qquad \tilde{G}(p) = \int d^4 x \, e^{i p \cdot x} G(x)$$

Scansionato con CamScanner

PERTURBATIVE EXPANSION OF GREE FUNCTION

INTERACTION PICTURE :

$$A_I = e^{iH_0(t-t_0)} A_S(t_0) e^{-iH_0(t-t_0)}$$

$$|\psi_t\rangle_I = e^{iH_0(t-t_0)} |\psi_{st}\rangle_S = e^{iH_0(\,)} e^{-iH(\,)} |\psi, t_0\rangle$$

$$\phi(t\vec{x}) = \underbrace{e^{iH(t-t_0)} e^{-iH_0(t-t_0)}}_{U^+(t,t_0)} \quad \phi_I(t,\vec{x}) \underbrace{e^{+iH_0(t-t_0)} e^{-iH(t-t_0)}}_{U(t,t_0)}$$

PROPERTY OF U :

$U^+(t,t_0) = U(t_0,t)$ → hence $U = e^{iH_0(\,)} e^{-iH(\,)} = e^{-iH(\,)} e^{iH_0}$

I can switch $\exp(H_0)$ and $\exp(H)$

$$U(t_2,t_1)U(t_1,t_0) = U(t_2,t_0)$$

$$i\frac{\partial U}{\partial t} = H_I(t)\, U(t,t_0) \qquad H_I(t) = e^{iH_0(t-t_0)} H_{int}(t) e^{-iH_0(t-t_0)}$$

$$U(t,t_0) = T\left\{ e^{-i\int_{t_0}^{t} dt' H_I(t')} \right\} = T\left\{ e^{-i\int d^4x' \mathcal{H}_I(x')} \right\}$$

GREN FUNCTION

$$\langle \Omega | T\{\phi(x_0)\cdots\phi(x_n)\}|\Omega\rangle = \frac{\langle 0| T\{\phi_I(x_1)\cdots\phi_I(x_n) e^{-i\int_{-\infty}^{\infty} d^4x' \mathcal{H}_I(x')}\}|0\rangle}{\langle 0| T\{ e^{-i\int_{-\infty}^{\infty} d^4x' \mathcal{H}_I(x')}\}|0\rangle}$$

IMP!

$$-i\int d^4x' \mathcal{H}_I(t',\vec{x}') = -i\int d^4x' \left(e^{-iH_0(t'-t_0)} \mathcal{H}_{int}(x') e^{iH_0(t'-t_0)} \right)$$

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PERTURBATIVE EXPANSION OF GREEN FUNCTION

(1) relating to ϕ_I

$$\langle \Omega | \phi_1 \phi_2 | \Omega \rangle \quad \downarrow \quad U_{t_0}^{\dagger} \phi_{1I} \; {}_{t_0} = \phi_1$$

$$\langle \Omega | U_{t_0}^{\dagger} \phi_1 U_{t_2} \phi_2 U_{t_0} | \Omega \rangle \quad \downarrow \quad t \gg t_1 > t_2 \gg -t$$

$$\langle \Omega | U_{t_0}^{\dagger} \, T\{ \phi_1 \phi_2 \, U(t,-t) \} \, U_{-t_0} | \Omega \rangle$$

(2) $|\Omega\rangle$ to $|0\rangle$

$$|0(t)\rangle = e^{-iHt}|0\rangle = e^{-iE_0 t}\langle \Omega|0\rangle|\Omega\rangle + \sum_{n=1}^{\infty} e^{-iE_n t}\langle n|0\rangle|n\rangle$$

- taking $t \to \infty(1-i\alpha)$ only Vacuum remains

$$|0(t)\rangle \underset{t\to\infty(1-i\alpha)}{\simeq} e^{-iE_0 t}\langle \Omega|0\rangle|\Omega\rangle$$

$$|\Omega\rangle = \lim_{t\to\infty(1-i\alpha)} \frac{1}{e^{-iE_0 t}\langle \Omega|0\rangle} e^{-iHt}|0\rangle$$

- Restory $t+t_0$

$$|\Omega\rangle = \lim_{\text{//}} \frac{1}{e^{-iE_0(t+t_0)}\langle \Omega|0\rangle} e^{-iH(t+t_0)}|0\rangle$$

- adding $+iH_0(t+t_0)$

$$|\Omega\rangle = \lim_{\text{//}} \frac{1}{\text{///}} \underbrace{e^{-iH(t+t_0)} e^{iH_0(t+t_0)}}_{U(t_0-t)} |0\rangle$$

$$U(-t,t_0)|\Omega\rangle = \lim_{t\to\infty(1-i\alpha)} \frac{1}{e^{-iE_0(t+t_0)}\langle \Omega|0\rangle}|0\rangle \qquad \& \quad \langle \Omega|U^{\dagger}(t,t_0) = \lim_{t\to\infty(1-i\alpha)} \frac{1}{e^{-iE_0(t-t_0)}\langle 0|\Omega\rangle}|0\rangle$$

(3) Green to $|0\rangle$

- $$\langle \Omega | \phi\phi | \Omega \rangle = \lim_{t\to\infty(1-i\alpha)} \boxed{\frac{1}{|\langle 0|\Omega\rangle|^2 e^{-iE_0 t}}}^{A} \langle 0 | T\{ \phi \phi \, U(t,-t)\} |0\rangle$$

- A

$$1 = \langle \Omega|\Omega\rangle = \langle \Omega | U_{t_0}^{\dagger} U_{t_0} U_{-t_0}^{\dagger} U_{-t_0} | \Omega \rangle = A \langle 0 | U_{t_0} U_{-t_0}^{\dagger} |0\rangle$$

$$A = \frac{1}{\langle 0 | U_{t-t} |0\rangle}$$

$$-i\int_{-\infty}^{+\infty} dx \, \mathcal{H}_I$$

- $$\langle \Omega | \phi\phi | \Omega \rangle = \lim_{t\to\infty} \frac{\langle 0|T\{\phi\phi \, U(t-t)\}|0\rangle}{\langle 0|U(t-t)|0\rangle} = \frac{\langle 0|T\{\phi\phi \, e^{-i\int_{-\infty}^{\infty} dx \, \mathcal{H}_I}\}|0\rangle}{\langle 0|T\{e^{-i\int_{-\infty}^{\infty} dx \, \mathcal{H}_I}\}|0\rangle}$$

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S MATRIX AND FEYNMANN RULES VIA WICK'S THEOREM

$$S_{kp} = \langle k | \lim_{\Delta T \to \infty} e^{-iH\Delta T} | p \rangle = \langle k | T\{ e^{-i\int_{-\infty}^{+\infty} dx^4 H_I(x)} \} | p \rangle$$

H_I → Hamiltoniano di interazione

poiché

$$S = T\{ e^{-i\int dx^4 H_I(x)} \}$$

Espando in serie di Dyson

$$S = \mathbb{1} + (-i)\int dx^4 T\{H_1(x)\} + \frac{(-i)^2}{2!}\int dx^4 \int dy^4 \, T\{H_I(x) H_I(y)\} + \dots$$

$$S = \mathbb{1} + \underbrace{\qquad\qquad\qquad\qquad\qquad\qquad}_{iT}$$

A noi interessa il contributo di iT poiché presupponiamo $|k\rangle \neq |p\rangle$ poiché $\langle k|\mathbb{1}|p\rangle = \langle k|p\rangle = 0$ diversi, dan contributo nullo

$$S_{kp} = \cancel{\langle k|p\rangle}_{\text{se } k \neq p} + (-i)\int dx^4 \langle k | T\{H_I(x)\} | p\rangle + \frac{(-i)^2}{2!}\int dx^4 \int dy^4 \langle k | T\{H_I(x) H_I(y)\} | p\rangle + \dots$$

ora usiamo il teorema di Wick's per ridurre $T\{H_I\}$ ad un prodotto di contrazioni tra i campi dentro H_I e poi contraiamo i campi rimasti con le particelle iniziali e finali $\langle k|$ e $|p\rangle$

- contrazione interne → danno propagatori
- ~~contrazioni con $|k\rangle$ e $|p\rangle$~~ → danno external legs

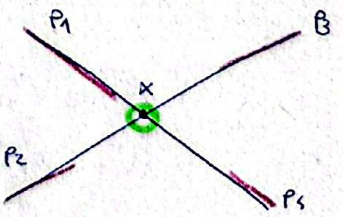
SCALAR FIELD

Considero $\phi(p_1)\phi(p_2) \to \phi(p_3)\phi(p_4)$ con $p_1 \neq p_2 \neq p_3 \neq p_4$ $\quad H_I = \frac{\lambda}{4!}\phi^4$

$$S_{kp} = (-i)\frac{\lambda}{4!}\int dx^4 \langle p_3 p_4 | \phi(x)\phi(x)\phi(x)\phi(x) | p_1 p_2\rangle \times 4! \text{ possibili contrazioni diverse}$$

1° ordine

$$S_{kp} = -i\lambda \frac{4!}{4!} \cdot 1 \cdot 1 \cdot 1 \cdot 1 = -i\lambda$$



⇒ solo external legs! È il VERTICE!

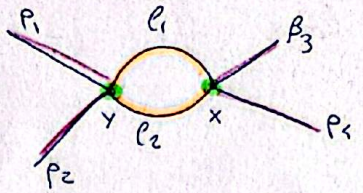
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Sup 2° ordine $= \frac{(-i\lambda)^2}{2!(4!)^2} \int dx^4 \int dy^4 \langle p_3 p_4 | \phi_x \phi_x \phi_x \phi_x \phi_y \phi_y \phi_y \phi_y | p_1 p_2 \rangle$

ora possibilità di contrazione = 1 loop diagram

$\Rightarrow \frac{(-i\lambda)^2}{2!(4!)^2} \cdot 4! \cdot 4! \int dx dy D(x-y)^2$

↑ quando esplicito LSZ otteng. Fourier di $\tilde{D}(e_1)$



Regole di Feynmann for scalars

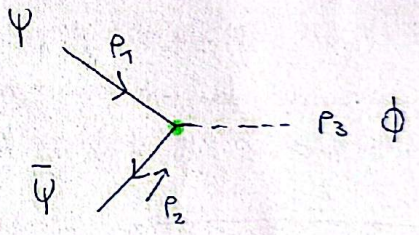
- PROPAGATORS = $\tilde{D}(e) = \frac{i}{p^2 - m^2}$
- EXTERNAL LEGS = $\phi |p\rangle = 1$
- VERTEX = $(-i\lambda)$ (1° ord diagram )

FERMIONS

Considero $\psi(p_3)\bar{\psi}(p_2) \longrightarrow \psi(p_3)\psi(p_4)$ con Yukawa $H_1 = g\bar{\psi}\psi\phi$

1° ordine = non esiste un grafico con solo un vertice per tale processo ... esiste però per un processo che ho $\bar{\psi}(p_1)\psi(p_2) \to \phi(p_3)$ due fermioni e uno scalare finale ... ci vedi il vertice stesso!

VERTICE = $(-ig) \int dx^4 \langle p_3 | \bar{\psi}(x)\psi(x)\phi(x) | p_1 p_2 \rangle$



$\Rightarrow (-ig) \cdot 1 \cdot \bar{V}^s(p_1)\, U^s(p_4)$

1 di ϕ; $\bar{V}^s(p_1)$ di $\bar{\psi}$; $U^s(p_4)$ di ψ

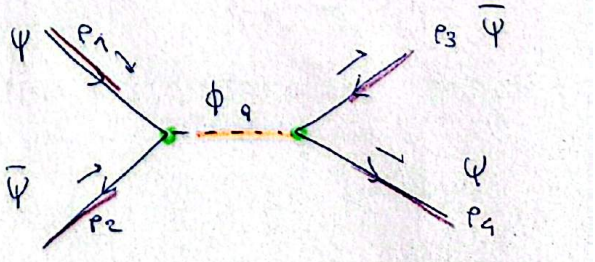
$\Rightarrow$ Sono tutte ext legs!

2 ordine $= \frac{(-ig)^2}{2!} \int dx^4 \int dy^4 \langle p_3 p_4 | \bar{\psi}(x)\psi(x)\phi_x \bar{\psi}_y \psi_y \phi_y | p_1 p_2 \rangle \times 2!$

prodi potrei contrarre sia così che scambiando x e y

$\Rightarrow (-ig)^2 \bar{U}^s(p_4) V^s(p_3) D(q) \bar{V}^r(p_2) U^r(p_1)$

($\bar{U}$: ψ; V: $\bar{\psi}$; $\bar{V}$: $\bar{\psi}$; U: ψ)



NOTA che metto primo sempre quello della freccia finale

legge così ↘ ↙ legge così

FEYNMANN RULES FOR FERMIONS

- PROPAGATORS $\tilde{S}(p) = \frac{i(\not{p}+m)}{p^2-m^2}$
- EXT LEGS $\overline{\psi|p\rangle} = \bar{v}^s(p)$ $\quad \langle k|\overline{\psi} = v^s(k)$ (initial)
  $\overline{\psi|p\rangle} = u^s(p)$ $\quad \langle k|\overline{\psi} = \bar{u}^s(k)$ (final)
- PROPAGATORS $(-ig)$ for vertex ψ, $\bar{\psi}$, ϕ ...

PROPAGATORS

Si ottengono dalla contrazione $\overline{\psi_x \bar{\psi}_y} = D(x-y)$ e poi calcolato sotto $\int dx^4$ integrale di Fourier $\int dz\, e^{ipz}$ mi do $D(x-y) \to \tilde{D}(p)$ in momento space

EXT LEGS

SCALAR

$\phi|q\rangle = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_p}} \left(a_p e^{-ipx} + a_p^\dagger e^{ipx}\right)|q\rangle = \int \frac{d^3p}{(2\pi)^3} \left(a_p e^{-ipx} + a_p^\dagger e^{ipx}\right) \sqrt{2E_q}\, a_q^\dagger |0\rangle =$ (crea una particella in più creando il vuoto come)

$= \int \frac{d^3p}{(2\pi)^3} a_p a_q^\dagger e^{-ipx} \frac{\sqrt{2E_q}}{\sqrt{2E_p}} |0\rangle = \int \frac{dp^3}{(2\pi)^3} \left([a_p a_p^\dagger] + a_p^\dagger a_p\right) \frac{\sqrt{E_q}}{\sqrt{E_p}} e^{-ipx}|0\rangle =$

$= \int \frac{dp^3}{(2\pi)^3} (2\pi)^3 \delta^{(3)}(p-q) \frac{\sqrt{2E_q}}{\sqrt{2E_p}} e^{-ipx}|0\rangle = 1\, e^{-ipx}|0\rangle$

$\to \phi|q\rangle = e^{-ip\cdot x}|0\rangle$ poi il fattore $e^{-ip\cdot x}$ si elimina quando vado in Fourier trasf. (crea un antiparticella in x)

FERMION

$\psi(x)|q,s\rangle = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_p}} \sum_r \left(a_p^r u_p^r e^{-ipx} + b_p^{r\dagger} v_p^r e^{ipx}\right) \sqrt{2E_q}\, a_p^{s\dagger}|0\rangle =$

(particella creata con a e non con b)

$= \int \frac{d^3p}{(2\pi)^3} \sqrt{\frac{2E_q}{2E_p}} \sum_r \left(\{a_p^r, a_q^{\dagger s}\} + a_q^{\dagger s} a_p^r\right) u_p^r e^{-ipx}|0\rangle =$

$= \int \frac{d^3p}{(2\pi)^3} \sqrt{\frac{2E_q}{2E_p}} (2\pi)^3 \sum_r \delta_{rs}\, \delta^{(3)}(p-q)\, e^{-ip\cdot x} u_p^r |0\rangle =$

$= \int \frac{d^3p}{(2\pi)^3} (2\pi)^3 \sqrt{\frac{E_q}{E_p}}\, \delta^{(3)}(p-q)\, e^{-ip\cdot x} u_p^s |0\rangle = u_q^s e^{-iqx}|0\rangle$

$\psi|q,s\rangle = u_q^s e^{-iq\cdot s}|0\rangle$ $\to$ il termine $e^{-iq\cdot s}$ si elimina in Fourier tx

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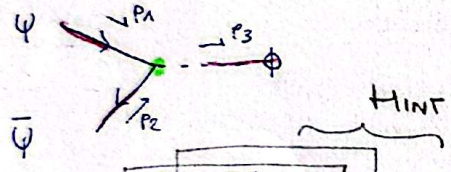
VERTEX

comes from coefficient in the expansion at 1° order, not considering external legs. È tutto ciò che rimane dal 1° order senza external legs.

Cioè lo calcolo prendo il diagramma del vertice ⟩··· e usando le procedure appena visto $-i\int d^4 \langle k|H_I|q\rangle$

- ≠ contrazioni equivalenti per quel digramma
- coefficiente di coupling
- $-i$ per 1° st order
- $\partial_\mu\phi$ incog $\longrightarrow -iP_\mu$
  $\partial_\mu\phi$ outgog $\longrightarrow iP_\mu$
- $A_\mu A^\mu \longrightarrow g_{\mu\nu}$ need to lower all indices

VIA LSZ FORMULA



① $\langle 0|\{\psi_1\bar\psi_2\phi_3\int dz^4\,\bar\psi_z\psi_z\phi_z\}|0\rangle(-ig)$

START on the vacuum and ext legs fields explicitate

② $\int dz^4\, S(x_1-z)S(z-x_2)D(x_3-z)\quad(-ig)$

Contrai con Wick's, ricordando di swappare e prendere un meno per ogni fermione che swappano. Devo fare contratti $\psi\bar\psi$ e MAI $\psi\psi$ perdi sono diversi!

③ FACCIO LO TRASF di Fourier

$\tilde S(p_1)\,\tilde S(p_2)\,\tilde D(p_3)\quad(-ig)\quad(2\pi)^4\delta^4(p_1+p_2-p_3)$

prop delle ext legs · vertice · overall momento conservato

④ Confronto con LSZ e le ext legs diventano le ext legs di Feynman, si tolgono i loro propagatori

$U^{s1}(p_1)\,\bar V^{s2}(p_2)\quad(-ig)\quad(2\pi)^4\delta^4(p_1+p_2-p_3) = \langle 1|S|p\rangle$

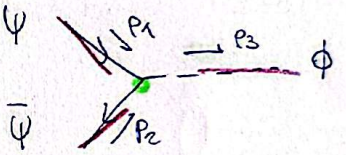
$\langle 1|S|p\rangle =$

⑤ Confronto con $i\mathcal{M}\,(2\pi)^4\delta^4(p_1+p_2-p_3)$ e ricavo $i\mathcal{M}$

$i\mathcal{M} = U^{s1}(p_1)\,\bar V^{s2}(p_2)\quad(-ig)$

ext legs · vertex

STRAIGHTFORWARD



① $(-i)\langle p_3|\{\int dz^4\, H_{int}\}|p_1p_2\rangle =$

$= \langle p_3|\{\int dz^4\,\bar\psi_z\psi_z\phi_z\}|p_1p_2\rangle(-ig)$

Start with expression of S matrix between initial and final states

② Contract with Wick's with external legs

$(-ig)\langle p_3|\{\int dz^4\,\bar\psi\,\psi\,\phi_z\}|p_1p_2\rangle$

remember to swap when necessary for minus sign

③ This already gives ext legs when doing FOURIER TRANSform

$U^{s1}(p_2)\,\bar V^{s2}(p_1)\quad(-ig)\quad(2\pi)^4\delta^4(p_1+p_2-p_3)$

ext legs · overall momento of the vertex

④ Confronto con $i\mathcal{M}(2\pi)^4\delta^4(p_1+p_2-p_3)$

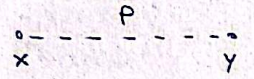
$i\mathcal{M} = U^{s2}(p_2)\,\bar V^{s2}(p_2)\quad(-ig)$

ext leg · vertex

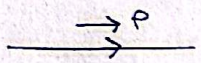
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FEYNMANN RULES

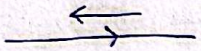
PROPAGATORS



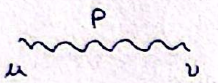
$\phi \qquad \tilde{D}(p) = \frac{i}{p^2 - m^2 + i\varepsilon}$



$\psi \qquad \tilde{S}(p) = \frac{i(\not{p} + m)}{(p^2 - m^2 + i\varepsilon)}$



$\bar{\psi} \qquad \tilde{S}(p) = \frac{i(-\not{p} + m)}{(p^2 - m^2 + i\varepsilon)}$



$A_\mu \qquad \tilde{D}_{\mu\nu}(p) = \frac{-ig^{\mu\nu}}{p^2 + i\varepsilon}$

NOTE

$$D(x-y) = \int \frac{d^4p}{(2\pi)^4} e^{-ip(x-y)} \tilde{D}(p)$$

$$\tilde{D}(p) = \int d^4x \, e^{ip\cdot x} D(x)$$

MA

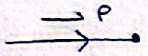
$$(2\pi)^3 \delta(p-k) = \int d^3x \, e^{-i(p-k)x}$$

$$(2\pi)^3 \delta(x-y) = \int dp \, e^{-ip(x-y)}$$

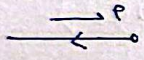
EXT LEGS



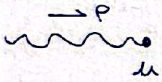
$= 1$



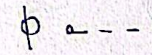
$\psi \qquad U^s(p)$



$\bar{\psi} \qquad \bar{V}^s(p)$



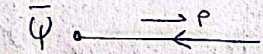
$A_\mu \qquad \varepsilon_\mu(p,\lambda)$



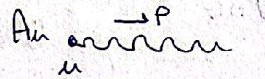
$\phi \qquad 1$



$\psi \qquad \bar{U}^s(p)$



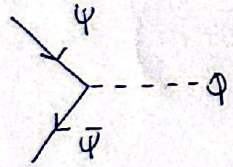
$\bar{\psi} \qquad V^s(p)$



$A_\mu \qquad \varepsilon_\mu^*(p,\lambda)$

NB $\mathcal{L}_{int} = -\mathcal{H}_{int}$
poi $(-i)$ se $\mathcal{H}_{int}$
$(+i)$ se $\mathcal{L}_{int}$

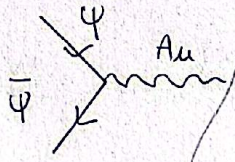
VERTICES



$\mathcal{L} = g\bar{\psi}\phi\psi$

$\downarrow$

$(-ig)$



$\mathcal{L} = g\bar{\psi}\gamma^\mu\psi A_\mu$

$\downarrow$

$(ig\gamma^\mu)|Q|$ ← carica fermione



$\mathcal{L} = \frac{\lambda}{4!}\phi^4$) moltiplicato di 4!

$\downarrow$

$(-i\lambda)$

Feynman Rule =

$(+i)^n \# \frac{g^m}{n!}$ → g is the coupling constant in $\mathcal{L}$

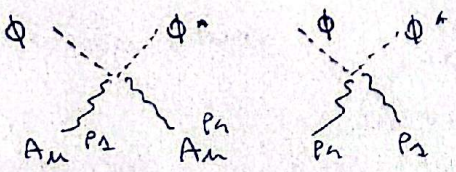
→ order of the diagram (= n° of vertices)

number of contractions equivalent

$$\mathcal{L} = e^2 A^\mu A_\mu \phi^* \phi \rightarrow ie^2 A_\nu A_\mu g^{\mu\nu} \phi^* \phi \rightarrow 2ie^2 g^{\mu\nu}$$

moltiplicato di due diagrammi

↓ dallo moltiplicato



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General rules for vertex contributions

→ se considero $\mathcal{L}_{int}$

- add i (or $-i$, a secondo della convenzione), se considero $-\mathcal{H}int$
- ϕ, ϕ^*, A_μ (with low index), $\psi, \bar{\psi}$ accounts for 1
- $\partial_\mu\phi$ incoming $\to -ip_\mu$ $\quad$ $\partial_\mu\phi^*$ incoming $\to -ip_\mu$
  $\partial_\mu\phi$ outgoing $\to +ip_\mu$ $\quad$ $\partial_\mu\phi^*$ outgoing $\to +ip_\mu$
- $A^\mu A_\mu \to A_\mu A_\nu g^{\mu\nu} \to g^{\mu\nu}$ I need to lower all A_μ indeces = gain $g^{\mu\nu}$

INCOMING

$$\overbrace{\partial^\mu\phi|p\rangle} = -ip^\mu e^{-ip\cdot x}|0\rangle \quad \left.\right\} \text{INCOMING}$$
$$\overbrace{\partial^\mu\phi^*|p\rangle} = -ip^\mu e^{-ip\cdot x}|0\rangle$$

perché $\phi = \int \frac{dp}{(2\pi)^3}\frac{1}{\sqrt{2}}\left(a_p e^{-ip\cdot x} + b_p^\dagger e^{ip\cdot x}\right)$

$\phi^* = \int \frac{dp}{(2\pi)^3}\frac{1}{\sqrt{2}}\left(a_p^\dagger e^{ip\cdot x} + b_p e^{-ip\cdot x}\right)$

$\Rightarrow$ per gli incoming $|p\rangle$ contraggo con particelle mancanti e quindi contraggo gli annichilatori a_p e b_p, i creatori $a_p^\dagger$ e $b_p^\dagger$ DANNO NOCCIO!

OUTGOING

$$\langle k|\overbrace{\partial^\mu\phi} = ik^\mu e^{ik\cdot x}\langle 0| \quad \left.\right\} \text{OUTGOING}$$
$$\langle k|\overbrace{\partial^\mu\phi^*} = ik^\mu e^{ik\cdot x}\langle 0|$$

perché $\partial\phi = \int \frac{dp}{(2\pi)^3}\frac{1}{\sqrt{2}}\left(a_k(-ik)e^{-ik\cdot x} + b_k^\dagger(ik)e^{ik\cdot x}\right)$

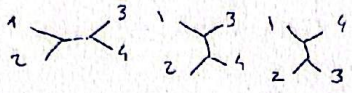
$\partial^\mu\phi^* = \int \frac{du}{(2\pi)^3}\frac{1}{\sqrt{2}}\left(a_u^\dagger(iu)e^{iu\cdot x} + b_u e^{-iu\cdot x}(-iu)\right)$

$\Rightarrow$ per gli outgoing contraggo $\langle k|$ con i creatori $a^\dagger$ e $b^\dagger$!!

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CALCOLARE L'ELEMENTO DI MATRICE

1) Regole di Feynmann
   - propagatori
   - ext lines
   - vertices ⟶ convenzione $-i(\dots)$ se uso $\int dx^3 \mathcal{H}_{int}$ $\quad +i(\dots)$ se uso $\int dx^3 \mathcal{L}_{int}$ $\quad$ perché $\mathcal{L}_{int} = -\mathcal{H}_{int}$
   - moltiplicate

2) Mi viene dato un processo, scrivere tutti i possibili diagrammi per quel processo
   - S channel t channel U channel 
   - molteplicata ~~$\psi\bar{\psi} \to \psi\bar{\psi}$~~ ⇒ 4 possibilità

3) Scrivere l'elemento di matrice non polarizzato (cioè mediato sugli spin iniziali e sommato sui finali)

SEGUIRE LA LINEA FERMIONICA AL CONTRARIO



Fermioni $A\alpha B\beta C$ vertici

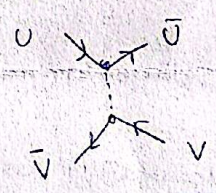
a) $M_{TOT} = M_U + M_t + M_S$

$M_U =$ } = regole di Feynmann
$M_t =$
$M_S =$

TIP: Scrivere la matrice partendo dal vertice di dx e poi di sx oppure dal basso verso l'alto, ricordando di propagatori in mezzo e facendo sì di avere sempre $\bar{U}U$ e $\bar{V}V$

$\bar{V} g V \quad \frac{i}{(q^2 - M^2)} \quad \bar{U} g U$

vertice bosone — propagatore — vertice altro



b) $M_U^\dagger =$ } TIP: $(ABC)^\dagger = C^\dagger B^\dagger A^\dagger$ $\quad \gamma^{\mu\dagger} = \gamma^0 \gamma^\mu \gamma^0$ $\quad \gamma^0 \gamma^5$ anti commutano
$M_t^\dagger =$
$M_S^\dagger =$

c) $|M_{TOT}|^2 = (M_U + M_t + M_S)^\dagger (M_U + M_t + M_S) = |M_S|^2 + |M_U|^2 + |M_t|^2 + M_t^\dagger M_U + M_U^\dagger M_t + M_t^\dagger M_S + M_S^\dagger M_t + M_S^\dagger M_U + M_U^\dagger M_S$

d) mediare sugli spin iniziali s_1 s_2 e sommare sui finali s_3 s_4

$$\frac{1}{2}\sum_{s_1}\frac{1}{2}\sum_{s_2}\sum_{s_3}\sum_{s_4}|M_{TOT}|^2$$

- Strutturare scrivere i prodotti con i spinori numeri $\quad \bar{V}V = \bar{V}^\alpha V_\alpha \dots$
- riconoscere $\sum_s \bar{V}^s V^s = \not{p} - m \quad \sum_s U^s \bar{U}^s = \not{p} + m$
- riconoscere le Tracce



due tracce — una traccia

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=> sfruttare $Tr[\gamma^\alpha \gamma^\beta \gamma^\gamma \gamma^\delta] = 4[g^{\alpha\beta} g^{\gamma\delta} - g^{\alpha\gamma} g^{\beta\delta} + g^{\alpha\delta} g^{\beta\gamma}]$

$Tr[\gamma^\mu \gamma^\nu] = 4 g^{\mu\nu}$

$Tr[\text{dispari } \gamma] = 0$

e) Riscrivere tutto con Mandelstam

$$u = (p_1 - p_4)^2 = -2p_1 \cdot p_4 + p_1^2 + p_4^2 = (p_2 - p_3)^2 = p_2^2 + p_3^2 - 2p_2 \cdot p_3$$

$$t = (p_1 - p_3)^2 = -2p_1 \cdot p_3 + p_1^2 + p_3^2 = (p_2 - p_4)^2 = p_2^2 + p_4^2 - 2p_2 \cdot p_4$$

$$s = (p_1 + p_2)^2 = p_1^2 + p_2^2 + 2p_1 \cdot p_2 = (p_3 + p_4)^2 = p_3^2 + p_4^2 + 2p_3 \cdot p_4$$

PROPRIETÀ'

$s + t + u = \sum_i m_i^2$ i indice che scorre su tutte le particelle iniziali e finali

se alte energie $p_i \gg m_i$

$s + t + u = 0$

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COUNTING DIVERGENCIES

$\frac{\phi^n}{n!}$ Interacting theories

$$D = dL - 2P$$

$$D = d + V\left(n\frac{(d-2)}{2} - d\right) - N\left(\frac{d-2}{2}\right)$$

d = dim dello spazio L = # loops P = # propagators

V = # vertici n = grado della teoria

N = ext scalar legs

Se $D = 0$ diverge come $\sim \log(\Lambda)$

$D > 0$ div come $\sim \Lambda^D$

$D < 0$ non diverge

Se $n\frac{(d-2)}{2} - d < 0$ Super renormalizable theory

$n\frac{(d-2)}{2} - d > 0$ NON renormalizable

$n\frac{(d-2)}{2} - d = 0$ Renormalizable

$\frac{\phi^4}{4!}$ interaction and Fermion scalar interaction

$$D = 4L - 2P_S - P_f$$

P_S = # prop scalari P_f = # propagatori fermioni

$$D = 4L - N_S - \frac{3}{2}N_f$$

N_S = ext leg scalari N_f = ext leg fermioni

In generale

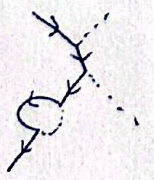
$$\mathcal{M} \sim \prod_L \int \frac{d^4k_L}{(2\pi)^4} \left(\frac{i}{k^2 - m_\phi^2}\right)^{P_S} \left(\frac{i\not{k} - m_\psi}{k^2 - m_\psi^2}\right)^{P_f} \left(\frac{-i\, g_{em}}{k^2}\right)^{P_{phot}}$$

$\sim k^{-2}$ $\sim k^{-1}$ $\sim k^{-2}$

$$4L - 2P_S - P_f - 2P_{phot}$$

! EVEN IF OVERALL D totale del diagramma è > 0 ci possono essere SUBDIAGRAMS DIVERGENTI!

EX:



$L = 1$
$P_S = 1$
$P_f = 3$
$D_{tot} = 4 \cdot 1 - 2 \cdot 1 - 3 = -1$
NON DIVERGE

TUTTAVIA IL SUB DIAGRAM:



$L = 1$
$P_S = 1$
$P_f = 1$
$D_{sub} = 4 \cdot 1 - 2 \cdot 1 - 1 = 1$
DIVERGE!!

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DISCUSSING RENORMALIZATION

$$S_{\text{Action}} = \int dx^d \mathcal{L} \qquad d = \text{dim of space}$$

Since $[\mathcal{L}] = [m]^d$ and $[x] = \frac{1}{[m]}$ in natural unit then $\Rightarrow [S] = [m]^0$

$$[\partial_x] = [x]^{-1} = [m]$$

Dimension of ϕ

$$\mathcal{L} = m^2 \phi^2 + \dots$$

$$[\mathcal{L}] = [m]^2 [\phi]^2 \rightarrow [m]^{d-2} = [\phi]^2 \rightarrow [\phi] = [m]^{\frac{d-2}{2}}$$

Dimension of ψ

$$\mathcal{L} = \sim m \bar{\psi}\psi$$

$$[m]^d = [m][\bar{\psi}\psi] \longrightarrow [\bar{\psi}\psi] = [m]^{d-1}$$

Dimension of A_μ

$$\mathcal{L} = -\frac{1}{4} F^{\mu\nu} F_{\mu\nu} = -\frac{1}{4} (\partial_\mu A_\nu - \partial_\nu A_\mu)(\partial^\mu A^\nu - \partial^\nu A^\mu) \sim (\partial_\mu A_\mu)^2$$

$$[m]^d = [m]^2 [A_\mu]^2 \rightarrow [A_\mu] = [m]^{\frac{d-2}{2}}$$

Dimension of couplings

$$g\, [\bar{\psi}\psi]^f [\phi]^s [A_\mu]^\gamma$$

dove f = # coppie fermioniche
ϕ = # scalari
γ = # fotoni

$$[g] = [m]^{d - f(d-1) - s\frac{(d-2)}{2} - \gamma\frac{(d-2)}{2}}$$

Se $\square > 0 \Rightarrow$ SUPER RENORMALIZABLE

$\square = 0 \Rightarrow$ RENORM

$\square < 0 \Rightarrow$ NON RENORM

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